Code No: 132AB **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year II Semester Examinations, March/April - 2023
MATHEMATICS – II
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) State the conditions for the existence of Laplace transform of a function. [2]
b) State the convolution theorem for Laplace transforms. [3]
c) Show that $\beta(m,n)=\beta(n,m)$. [2]
d) Show that $\Gamma(n+1)=n\Gamma(n)$. [3]
e) Evaluate $\int_0^4\int_0^{x^2} e^{y/x}dydx$. [2]
f) Evaluate $\iint xy\,dxdy$ over the positive quadrant of the circle $x^2+y^2=a^2$. [3]
g) Show that $\nabla r^n = nr^{n-2}R$ *where* $R=x\bar{i}+y\bar{j}+z\bar{k}$. [2]
h) If $uF=\nabla v$ where u,v are scalar fields *and* F is vector field, show that $F.curlF=0$. [3]
i) If $\bar{F}=3xy\bar{i}-y^2\bar{j}$, evaluate $\int \bar{F}.d\bar{R}$, where C is the curve in the xy – plane $y=2x^2$ from $(0,0)$ *to* $(1,2)$. [2]
j) Find the total work done by the force $\bar{F}=3xy\bar{i}-y\bar{j}+2zx\bar{k}$ in the moving particle around the circle $x^2+y^2=4$. [3]

PART - B

(50 Marks)

2.a) Find $L\left(\frac{Cos\sqrt{t}}{\sqrt{t}}\right)$.

b) Using convolution theorem find the inverse Laplace transform of $\frac{s^2}{(s^2+a^2)(s^2+b^2)}$. [5+5]

OR

3.a) Find the inverse Laplace transform of $\frac{1}{2}\log\left(\frac{s^2+b^2}{s^2+a^2}\right)$.

b) Using Laplace transforms, solve: $y''-3y'+2y=e^{3t}$ *with* $y(0)=1$ *and* $y'(0)=0$. [5+5]

4.a) Show that $\beta(m,n)=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

b) Evaluate $\beta\left(\frac{7}{2},\frac{-1}{2}\right)$. [5+5]

OR

5.a) Evaluate $\int_0^\infty e^{-ax}x^{m-1}\sin bx\,dx$ in terms of Gamma functions.

b) Evaluate $\Gamma\left(\frac{-7}{2}\right)$. [5+5]

6.a) By double integration find the area enclosed by the curves: $y=\frac{3x}{x^2+2}$ *and* $4y=x^2$.

b) Find the volume enclosed by the cylinders $x^2+y^2=2ax$ *and* $z^2=2ax$. [5+5]

OR

7.a) By double integration find the whole area of the curve $a^2x^2=y^3(2a-y)$.

b) Find the volume common to the cylinders $x^2+y^2=a^2$ *and* $x^2+z^2=a^2$. [5+5]

8.a) Show that $div(grad\,r^n)=\nabla^2 r^n=n(n+1)r^{n-2}$.

b) If r is the distance of a point (x,y,z) from the origin, prove that

$curl\left(K\times grad\frac{1}{r}\right)+grad\left(K.curl\frac{1}{r}\right)=0$ where K is the unit vector in the direction OZ.

[5+5]

OR

9.a) For a solenoidal vector $\bar{F}$, show that $curl\,curl\,curl\,curl\,\bar{F}=\nabla^4\bar{F}$.

b) If $f=(x^2+y^2+z^2)^{-n}$, find *div grad* f and determine n *if div grad* $f=0$. [5+5]

10.a) Evaluate $\int_S \bar{F}.\hat{n}\,ds$ where $\bar{F}=18z\bar{i}-12\bar{j}+3y\bar{k}$ and S is the portion of the plane $2x+3y+6z=-12$ in the first octant.

b) Verify the Greens Theorem for $\int_C\left[(3x-8y^2)dx+(4y-6xy)dy\right]$

where C is the boundary of the region bounded by $x=0, y=0$ *and* $x+y=1$. [5+5]

OR

11.a) Verify the Stokes Theorem for $\bar{F}=(y-z+2)\bar{i}+(yz+4)\bar{j}-xz\bar{k}$ where S is the surface of the cube $x=0, y=0, z=0, x=2, y=2, z=2$ above the xy-plane.

b) Verify Divergence Stokes Theorem for $\bar{F}=y\bar{i}+x\bar{j}+z^2\bar{k}$ over the cylindrical region bounded by $x^2+y^2=9,\ z=0$ *and* $z=2$. [5+5]

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